Plate sources & provenance

Every image used on a card records its source, credit line and licence here. `src-*` files are the originals (git-ignored, re-fetchable from the URL); `plate-*` files are the processed crops that go on the card.

Unless noted otherwise, every image below is a work of the U.S. Government (NASA) or a NASA / university partner (LROC is NASA / GSFC / Arizona State University) and is in the public domain, Wikimedia Commons tag `PD-USGov-NASA`. Batch-1 plates were retrieved 2026-09-01 from Wikimedia Commons via `Special:FilePath/<file name>`.

Processing tool: ImageMagick 7. Common steps: `-colorspace sRGB -strip -quality 82`.

L1 · Moon

plate-L1-scene.jpg (left / scene panel) ← `src-l1-scene-moon-mountains.jpg` - Full Moon rising over a snow-lit mountain ridge at sunset. A concept card, so the left panel carries an evocative scene instead of a locator map. - Supplied by Aaron from `/Volumes/Starfish_Backup/moon-field-card/` (filename marked "no-copyright"). **Attribution / licence not yet verified — confirm origin before the deck ships.** - Processing: resized to 1000 px wide, EXIF stripped, JPEG q82.

plate-L1-photo.jpg (right / close-up panel) ← `src-moon-nearside-lro.jpg` - LROC Wide Angle Camera near-side mosaic. Credit: NASA / GSFC / Arizona State University. - Commons: `File:Moon nearside LRO.jpg` - Processing: resized to 1200 px.

L2 · Earthshine

Batch-1 Apollo 12 plate replaced 2026-09-01 with two Aaron-supplied images.

plate-L2-scene.jpg (left / scene panel) ← `src-l2-scene-crescent-trees.jpeg` - Twilight crescent Moon low over a tree-line horizon, orange-to-blue sky. - Supplied by Aaron from `/Volumes/Starfish_Backup/moon-field-card/`. **Attribution / licence not verified** (same open question as L1). - **Low resolution: source is only 276 × 478 px.** Upscaled for the panel; fine for screen preview, not for the print PDF. Needs a larger version. - Processing: resized to 1000 px wide, EXIF stripped, JPEG q82.

plate-L2-photo.jpg (right / close-up panel) ← `src-l2-earthshine-crescent.png` - Telescopic crescent Moon with strong earthshine, whole disc visible in ashen light, bright sunlit limb. 1110 × 1043 source. - Supplied by Aaron from `/Volumes/Starfish_Backup/moon-field-card/`. **Attribution / licence not verified.** - Processing: resized to 1400 px wide, JPEG q82.

L3 · Mare / highland dichotomy

Panels reversed 2026-09-01 (Aaron): the annotated scope view is the left panel, the near/far two-up the right.

plate-L3-annotated.jpg (left panel) ← `src-l3-dichotomy-annotated.jpg` - Telescopic waning-gibbous Moon with two rings drawn on it: one over a smooth mare, one over the cratered southern highlands, showing the dichotomy in a single scope view. 700 × 887 source. - Supplied by Aaron from `/Volumes/Starfish_Backup/moon-field-card/`. **Attribution / licence not verified** (same open question as L1). - Processing: resized to 1200 px wide, JPEG q82.

plate-L3-nearfar.jpg (right panel) ← `src-moon-nearside-lro.jpg` + `src-moon-farside-lro.jpg` - Two-up: near side (left), far side (right), LROC WAC mosaics. The far side is almost all highland. Credit: NASA / GSFC / ASU. - Commons: `File:Moon nearside LRO.jpg`, `File:Moon Farside LRO.jpg` - Processing: each resized to 700 px, joined with a 14 px gutter (`magick +smush 14`).

L4 · Apennines

plate-L4-scene.jpg (left / scene panel) ← `src-l4-scene-apennines.png` - Telescopic view of the Apennine range along the terminator, Mare Imbrium to the west. 626 × 872 source. - Supplied by Aaron from `/Volumes/Starfish_Backup/moon-field-card/`. **Attribution / licence not verified.** A burned-in caption ("Raleigh - Stagecoach · April 10, 2008") points to a Raleigh, NC amateur capture; origin to confirm. - Processing: chopped 34 px off the top to remove the caption, resized to 1000 px wide, EXIF stripped, JPEG q82.

plate-L4-photo.jpg (right / close-up panel) ← `src-apenninus-lro.png` - LRO mosaic of the Montes Apenninus along the southeast rim of Mare Imbrium. Credit: NASA / GSFC / ASU. - Commons: `File:Montes Apenninus (LRO).png` - Processing: resized to 1500 px wide.

L5 · Copernicus

plate-L5-photo.jpg (front photo panel) ← `src-copernicus-lro-wac.png` - LROC Wide Angle Camera mosaic, 150 km across, north up. Credit: NASA / GSFC / Arizona State University. - Commons: `File:Copernicus (LRO) 2.png` (1264 × 1264, `PD-USGov-NASA`) - Processing: resized to 1400 × 1400. - Added 2026-09-01 as the straight-on primary view, replacing the Lunar Orbiter 2 oblique.

plate-L5-photo2.jpg ("Another view" inset) ← `src-lunarorbiter2-copernicus.jpg` - Lunar Orbiter 2, 1966, the "Picture of the Century" oblique. Credit: NASA / JPL / USGS. - Retrieved earlier from `https://science.nasa.gov/resource/copernicus-from-lunar-orbiter/` (asset `assets.science.nasa.gov/.../lunarorb_copernicus.jpg`). - Processing: trimmed ~70 px of top sky, 1400 px wide. - Was the front photo panel on the approved mockup; moved to the inset 2026-09-01.

plate-L5-locator.jpg (front locator panel) ← `src-lroc-wac-copernicus.png` - LROC WAC colour mosaic of Copernicus and its ray system ("Absolute Time", LROC featured image 480). Credit: NASA / GSFC / ASU. - Commons: `File:Absolute Time (LROC480 - copern lwac731a).png`, via `https://commons.wikimedia.org/wiki/Special:FilePath/Absolute%20Time%20(LROC480%20-%20copern%20lwac731a).png` - Processing: cropped off the burned caption strip, desaturated to ~24 %, 700 px wide, q84.

src-lroc238-copernicus-context.png / src-lroc473-copernicus-floor.png (not used) - LROC WAC context mosaic (annotated) and LROC NAC floor-detail mosaic. NASA / GSFC / ASU. - Kept as candidates for a wider production locator.

L6 · Tycho

Right panel changed 2026-09-01 (Aaron): the dramatic LRO NAC sunrise oblique is a spacecraft-only angle; replaced with a near-vertical view that matches what a telescope shows from Earth.

plate-L6-scene.jpg (left / scene panel) ← `src-l6-scene-tycho-wide.png` - Wide telescopic view of the southern highlands with Tycho and its rays. 411 × 495 source (low resolution; fine for screen, soft for print). - Supplied by Aaron from `/Volumes/Starfish_Backup/moon-field-card/`. **Attribution / licence not verified.** - Processing: resized to 1000 px wide, EXIF stripped, JPEG q82.

plate-L6-photo.jpg (right / close-up panel) ← `src-l6-tycho-lro-wac.png` - LROC Wide Angle Camera mosaic of Tycho, 120 km across, north up, near-vertical. Credit: NASA / GSFC / Arizona State University. - Commons: `File:Tycho LRO.png` (1058 × 1058, `PD-USGov-NASA`) - Processing: resized to 1400 × 1400, JPEG q82.

src-tycho-lro-centralpeak.jpg (not used) — `File:LRO Tycho Central Peak.jpg`, the LRO NAC sunrise oblique. Kept as a candidate for an "Another view" inset.

L7 · Altai Scarp (both panels re-pulled from LunaServ, 2026-09-02)

plate-L7-scene.jpg (left / locator) and **plate-L7-photo.jpg** (right / close-up) ← LROC QuickMap / LunaServ WMS, layer `luna_wac_global` (see the L20–L25 note below for the method). Centred on the scarp mid-arc (23.5°S, 24.5°E).

- Locator: 40° latitude field — Mare Nectaris (dark, NE) with Rupes Altai arcing round its southwest rim; ring `rx/ry 12`.
- Close-up: 13° field, framed to show the whole ~425 km arc of the scarp rather than one segment (the scarp's shape is its signature, and it is too subtle at moderate sun for a tight face view to read). `-normalize -sigmoidal-contrast 3x48%`.
- Credit: LROC WAC global mosaic, 100 m/px, via LROC QuickMap / LunaServ. NASA / GSFC / ASU.

- **Fixes the earlier bug:** the old right panel was `File:Rupes Altai - LROC - WAC.JPG` from the annotated `- LROC - WAC.JPG` Commons atlas set, which has burned-in white labels (Catharina, Fermat, Rupes Altai, …). Now unlabelled.
- Note: like all of Rupes Altai from an orthomosaic, this is honest but not dramatic; the scarp really comes alive only near the day 5–7 terminator.

plate-L7-photo2.jpg ("Another view" inset, label "At sunrise") ← `src-l7-photo2-altai-terminator.png` - **Aaron's own telescopic image** of Rupes Altai at the terminator: the scarp arc catching first light, deep shadow behind it, Piccolomini at the southern end. This is the dramatic low-sun look the orthomosaic cannot give. - Credit: Aaron Henderson, THG Media. - Processing: 4:3 crop `2100×1575+405+595` from the 2777 × 3434 original (settled between a too-far-south and a too-far-north attempt, per Aaron), resized to 1200 × 900, mild `-brightness-contrast 2x4`, quality 88. His natural tone kept (not desaturated).

L8 · Theophilus, Cyrillus, Catharina

plate-L8-scene.jpg (left / zoomed-out panel) ← `src-l8-scene-nectaris-lro.png` - The trio sits on the northwest rim of Mare Nectaris; this is a NW crop of the same LROC WAC basin mosaic used on L7. Credit: NASA / GSFC / ASU. - Commons: `File:Mare Nectaris (LRO).png` (2750 × 2750, `PD-USGov-NASA` family) - Processing: cropped to the NW quadrant (`-gravity northwest -crop 1950x2300+100+100`), resized to 1150 px wide, JPEG q82.

plate-L8-photo.jpg (right / overhead panel) ← `src-l8-scene-nectaris-lro.png` - All three craters from directly overhead: Theophilus (sharp, terraced, central peak) at top, Cyrillus (degraded) centre, Catharina (worn) bottom, the degradation sequence in one view. Cropped from the same LROC WAC basin mosaic as the left panel. Credit: NASA / GSFC / ASU. - Commons: `File:Mare Nectaris (LRO).png` (2750 × 2750, `PD-USGov-NASA` family) - Processing: cropped to the trio (`-crop 1120x1050+210+700`), resized to 1300 px wide, JPEG q82. Crop centring checked by eye against the rendered image. - Changed 2026-09-01 (Aaron): Apollo 16 oblique → overhead of the trio. An interim single-Theophilus WAC frame (`File:Central Peak Bedrock (LROC428 - Theophilus WAC).png`) was rejected for not showing all three.

Both L8 panels now derive from `File:Mare Nectaris (LRO).png` (different crops). Flag: find a distinct overhead if we want them from separate sources.

src-theophilus-as16.jpg (not used) — `File:Theophilus crater AS16-113-18295HR.jpg`, Apollo 16 oblique on Orion's approach. Kept as an "Another view" candidate.

L9 · Clavius

plate-L9-photo.jpg ← `src-clavius-lo4.png` - Lunar Orbiter 4 frame of Clavius and its curving chain of floor craters, 1967. Credit: NASA. - Commons: `File:Lunar Orbiter 4 FRAME 4107 M p13.png` - Processing: reduced to 8-bit, resized to 1300 px wide, centre-cropped to 1300 × 1300.

L10 · Mare Crisium (both panels, Earth-view render from LunaServ, 2026-09-02)

Both panels rebuilt as the **Earth / telescope view**, per Aaron: these are observing cards, and the Moon is never seen from directly overhead. Method: LunaServ WMS (`wms.im-ldi.com`, the LROC QuickMap engine), layer `luna_wac_global`, **orthographic** projection (`SRS=AUTO:42003,9001,<lon0>,<lat0>`) centred on a favourable-libration sub-observer point (7°E, 5°N for Crisium). BBOX in projection metres on the PROJ default sphere (radius 6 378 137). Feature projected coordinates computed and the window cropped to them; `-normalize -level …,1.3 -sigmoidal-contrast`.

- **plate-L10-photo.jpg** (right): ~1 500 km window on Mare Crisium — the foreshortened oval near the eastern limb, highland ring, floor wrinkle ridges, Picard and Peirce, the lunar edge curving down the right. The card's own description already notes this foreshortening; now the image matches.
- **plate-L10-scene.jpg** (left): wider Earth-view of the eastern near side, crop centred at projected `(4.4M, 1.5M)` so Crisium sits near centre with the limb curving down the right against **black sky** (WMS `BGCOLOR=0x000000` on both panels). `-gamma 1.5 -sigmoidal-contrast 1.4x38%`. Crisium ringed (`outline cx:56 cy:48 rx:6 ry:8`, drawn 2× loose by the renderer). Replaces Aaron's low-res supplied image and retires its identity-check flag.
- Credit: LROC WAC (LROC QuickMap / LunaServ), rendered as the Earth view. NASA / GSFC / ASU.
- Retires the letterboxed Apollo 17 mapping strip that was on the right.

Batch 2 (L11–L20)

LEFT panel = wide finder field; RIGHT panel = narrow eyepiece detail. All public-domain (NASA / LROC / Lunar Orbiter / Apollo). The Commons `… - LROC - WAC.JPG` atlas set was avoided where possible: it carries burned-in labels. Prose for these cards is not yet written.

> ***Superseded for L11–L19 (2026-09-02).*** *All of L11–L19, both panels, were rebuilt as the **Earth / telescope view** by `tools/build-plates-lunaserv.py` (see the L20–L25 section for the method: LunaServ `luna_wac_global`, orthographic `AUTO:42003` centred on a favourable-libration sub-observer point, black sky, then gamma/contrast). Locator = 44° field (crop eased ~15% toward disc centre), feature marked with an orange dot (`outline.cx/cy`); close-up = per-feature 3–20° field centred on the feature. Credits set to "LROC WAC (LROC QuickMap / LunaServ), rendered as the Earth view. NASA / GSFC / ASU." The per-card notes below describe the earlier public-domain plates and no longer match what ships.*

L9 · Clavius (left panel = VMA locator, 2026-09-01)

plate-L9-scene.jpg (left, locator map) ← `src-moonbase-wac.jpg` - ~50°-wide field centred on Clavius (58.8°S, 14.1°W), cropped from the project's Virtual Moon Atlas near-side base map. Maginus upper-left, the south-polar terrain along the bottom edge. The orange ring is drawn by the renderer from the card's `outline` field, not baked into the plate. - Source: **VMA `WAC_LOWSUN` level-2 tiles** (LROC WAC low-sun-angle mosaic, NASA / Arizona State University), assembled by `tools/build-moonbase.sh` into a 10000×5000 equirectangular map (`src-moonbase-wac.jpg`, git-ignored). - Processing: crop `2529x1389+3344+3439` from the base (longitude widened by 1/cos lat, then squeezed to `1100x1038` so craters read round), `-colorspace sRGB -brightness-contrast 3x5`, quality 88. Native grayscale kept. - This is the first card on the new plan: one VMA base map, every locator panel a crop of it. Supersedes the first-quarter telescopic crop (`File:First quarter moon.jpg`, Astrobond, CC BY-SA 4.0) which was itself a fix for the earlier Lunar Orbiter 4 frame. Both earlier `src-l9-*` files stay on disk (git-ignored), unused.

plate-L9-photo.jpg (right) ← `src-l9-narrow-clavius-lroc.jpg` - Clavius floor and its curving crater arc, LROC. NASA / GSFC / ASU. - Commons: `File:Clavius LROC.jpg` (707 × 758 — low resolution, soft for print). - Processing: resized to 1000 px wide.

L11 · Aristarchus

plate-L11-scene.jpg (left) ← `src-l11-wide-aristarchus-plateau-lro.png` - The Aristarchus Plateau with Vallis Schröteri; Aristarchus bright at lower right, Herodotus lava-flooded beside it. LROC WAC. NASA / GSFC / ASU. - Commons: `File:Aristarchus Plateau (LRO).png` (2706 × 2706, clean/unlabelled) - Processing: resized to 1300 px wide.

plate-L11-photo.jpg (right) ← `src-l11-narrow-aristarchus-as15.jpg` - Aristarchus, oblique close-up from Apollo 15. NASA. - Commons: `File:Aristarchus crater hrp162.jpg` (1437 × 1066) - Processing: resized to 1400 px wide.

L12 · Proclus

plate-L12-scene.jpg (left) ← `src-l12-wide-proclus-as17.jpg` - Proclus and the western shore of Mare Crisium, Apollo 17. NASA. - Commons: `File:Proclus crater AS17-150-23046.jpg` (1136 × 1136) - Processing: resized to 1200 px wide.

plate-L12-photo.jpg (right) ← `src-l12-narrow-proclus-as17.jpg` - Proclus interior, oblique from Apollo 17. NASA. - Commons: `File:Proclus crater hrp147.jpg` (800 × 1675) - Processing: centre-cropped to ~square, resized to 1200 px wide.

L13 · Gassendi

plate-L13-scene.jpg (left) ← `src-l13-wide-humorum-lro.png` - Mare Humorum, with Gassendi cut into its north rim. LROC WAC. NASA / GSFC / ASU. - Commons: `File:Mare Humorum (LRO).png` (2744 × 2744, clean/unlabelled) - Processing: resized to 1300 px wide.

plate-L13-photo.jpg (right) ← `src-l13-narrow-gassendi-lro.png` - Gassendi, LROC Wide Angle Camera. NASA / GSFC / ASU. - Commons: `File:Gassendi (LRO).png` (1520 × 1520) - Processing: resized to 1400 px wide.

L14 · Sinus Iridum

Both panels are crops of one NASA image (a public-domain LRO perspective view); flag for a distinct narrow view later.

plate-L14-scene.jpg (left) ← `src-l14-sinus-iridum-nasa.jpg` - Sinus Iridum and the Jura Mountains arc, LRO perspective view. NASA (`PD-USGov`). - Commons: `File:Sinus Iridum.jpg` (1428 × 788) - Processing: centre-cropped, resized to 1200 px wide.

plate-L14-photo.jpg (right) ← same source - Processing: tighter centre crop of the bay, resized to 1200 px wide.

L15 · Straight Wall (Rupes Recta)

plate-L15-scene.jpg (left) ← `src-l15-wide-thebit-lro.png` - The Straight Wall (thin dark line, centre) with Birt to its west and Thebit to its east, in eastern Mare Nubium. LRO. NASA / GSFC / ASU. - Commons: `File:Ancient Thebit (LRO).png` (2116 × 2116, clean/unlabelled) - Processing: resized to 1300 px wide.

plate-L15-photo.jpg (right) ← `src-l15-narrow-rupesrecta-as16.jpg` - Rupes Recta, Birt and Rima Birt, oblique from Apollo 16. NASA / GSFC / ASU. - Commons: `File:Rupes Recta Birt crater AS16-M-2486 ASU.jpg` (1600 × 1000) - Processing: resized to 1400 px wide.

L16–L100 · left (locator) panels — VMA base map, batch 2026-09-01

> ***Superseded (2026-09-02): all of L7, L10, L11–L100*** *rebuilt, both panels, as the* ***Earth / telescope view*** *by* `tools/build-plates-lunaserv.py` *(orthographic* `AUTO:42003` *from LunaServ* `luna_wac_global`*, favourable-libration sub-observer point, black sky, orange dot on the locator; locator 44° field, close-up per-feature 3–38°). Limb and polar features (Orientale, Peary, Drygalski, Leibnitz Mtns, Australe, Humboldtianum, Smythii, Humboldt, Inghirami, Marginis) render as the foreshortened limb slivers against black that they actually are from Earth. The nadir VMA / Trek description below now applies only to* ***L9****.*

`plate-L16-scene.jpg` … `plate-L100-scene.jpg` (85 files) generated by `tools/build-locator-plates.py 16 100`, each a crop of `src-moonbase-wac.jpg` (the assembled Virtual Moon Atlas `WAC_LOWSUN` near-side map — see the L9 entry).

- Centred on the feature's spine lat/lon; field of view scales with feature size (~36–64° of latitude); longitude widened by 1/cos(lat) then squeezed on export so craters read round. Output 1100 × 1038 (locator-pane aspect).
- `data/cards.js` for each of these got `locator`, `locatorCredit` ("LROC WAC low-sun mosaic (Virtual Moon Atlas). NASA / ASU.") and an `outline` ring sized from the feature's angular diameter.
- **Rougher on the equirectangular base (flagged, revisit):** L37, L56, L70, L73, L76, L80, L87, L88, L94, L96, L100 — high latitude or extreme limb, where the base map smears. L88 (Peary, 88.6°N) is the worst; its crop runs into the polar-cap tile edge.

L16–L100 · right (close-up) panels — LROC WAC mosaic, batch 2026-09-01

`plate-L16-photo.jpg` … `plate-L100-photo.jpg` (85 files) generated by `tools/build-closeup-plates.py 16 100`.

- Source: **LRO LROC WAC Global Mosaic, 100 m/px** (`LRO_WAC_Mosaic_Global_303ppd_v02`), the equirectangular WMTS pyramid served by **NASA Solar System Treks** (`trek.nasa.gov/tiles/Moon/EQ/...`, z0–z8). NASA / GSFC / Arizona State University; public domain.
- Per feature: zoom chosen so the field of view is ~1500 px; covering tiles fetched (cached under `.trekcache/`, git-ignored), stitched, cropped (longitude widened by 1/cos lat so the feature reads round), resized to 1400 × 1050, `-normalize -sigmoidal-contrast 3.5x46%`, quality 88.
- `data/cards.js` for each got `photo`, `photoTag:"NASA · LROC WAC"`, `photoCredit:"LROC WAC global mosaic (100 m/px). NASA / GSFC / Arizona State University."`
- **Poor — need a polar-stereographic source (equirect mosaic smears to streaks here): L88 Peary** (88.6°N), **L94 Drygalski** (79.3°S), **L96 Leibnitz Mountains** (85°S). Currently placeholder-grade.
- **Acceptable but wide / limb-distorted:** L56, L73, L80, L87, L100 — big basins or far-limb; framed as a representative chunk, not the whole feature.
- Big basins (L80 Orientale 930 km, L95 Procellarum ~2500 km, L54/L73/L77) are capped at a 9° field, so the close-up shows the centre, not the full ring system.

L20–L25 · both panels re-pulled from LROC QuickMap / LunaServ, 2026-09-02

`plate-L20..L25-{scene,photo}.jpg` rebuilt by `tools/build-plates-lunaserv.py`.

- Source: **`luna_wac_global`** ("LROC WAC Global 100 m/px"), served by the **LunaServ WMS** at `https://wms.im-ldi.com/` — the map engine behind LROC QuickMap (Space Exploration Resources / ASU). NASA / GSFC / Arizona State University; public domain.
- Direct `EPSG:4326` `GetMap` per panel, longitude span widened by 1/cos(lat) so the feature reads round; `-normalize -sigmoidal-contrast 3x48%`, quality 88. Locator panels 1100 × 1040 at a 40° latitude field; close-ups 1400 × 1050 at a per-feature field (2–7.5°). Raw fetches cached as `src-l2x-*-lunaserv.jpg` (git-ignored).
- `data/cards.js` for these six: `locatorCredit` / `photoCredit` set to "LROC WAC global mosaic, 100 m/px, via LROC QuickMap / LunaServ. NASA / GSFC / ASU.", `outline` ring re-sized to the new field.
- Noticeably sharper and better-toned than the NASA Trek pulls. LunaServ also serves polar-stereographic and per-point orthographic projections, so it is the intended fix for the 3 polar close-ups and the rough limb locators.

Production note

The mockup and batch-1 plates are public-domain stand-ins. The finished deck uses Aaron's own Seestar / processed captures in the close-up panel wherever one exists, with LRO / LROC / Lunar Orbiter / Apollo public-domain imagery as the fallback. The "Another view" inset is an optional second image per card; on L5 it carries the historic Lunar Orbiter 2 oblique.

The Copernicus locator labels and orange ring are placed by eye, not projected from coordinates; a production locator should use a wider projected mosaic.